> *restart*

Resistor values

- **Step 1 – Formula:**

$$V_{out} = V_{ref} \cdot \left(1 + \frac{R1}{R2}\right) \quad \Rightarrow \quad \frac{R1}{R2} = \frac{V_{out}}{V_{ref}} - 1$$

> $V_{out1} := 1.8 : V_{out2} := 3.3 : V_{in} := 4.5 :$

$V_{OUT} + 1\ V < V_{IN} < 16\ V$

> $V_{ref1} := 1.2 :$

> $solve\left(V_{out1} = V_{ref1} \cdot \left(R_{ratio} + 1\right), R_{ratio}\right)$

$$0.5000000000 \tag{1.1}$$

> $solve\left(V_{out2} = V_{ref1} \cdot \left(R_{ratio} + 1\right), R_{ratio}\right)$

$$1.750000000 \tag{1.2}$$

> $ratio1 := \frac{1}{2} : ratio2 := 1.75 :$

Ratio for 1V8: $\frac{R_1}{R_2} = \frac{1}{2}$

Ratio for 3V3: $\frac{R_1}{R_2} = 1.75$

Resistor package: 0603
Resistor series: E6/E12 series

<u>Resistor for 1V8:</u>

> $R_{2,1} := 68E+3 :$

> $R_{1,2} := ratio1 \cdot R_{2,1}$

$$R_{1,2} := 34000.00000 \tag{1.3}$$

$\boldsymbol{R_{1,1} = 33\ k\Omega}$
$\boldsymbol{R_{2,1} = 68\ k\Omega}$

R11:https://www.digikey.com/en/products/detail/panasonic-electronic-components/ERA-3AEB333V/1465890
R21:https://www.mouser.dk/ProductDetail/Vishay/MCT0603ME6802BP500?qs=sGAEpiMZZMtlubZbdhIBIE4nSA98ICq4y1fRBqNHL34%3D

> $R_{2,2} := 220E+3 :$

> $R_{1,2} := ratio2 \cdot R_{2,2}$

$$R_{1,2} := 385000. \quad (1.4)$$

$\boldsymbol{R_{1,2} = 39\ k\Omega}$
$\boldsymbol{R_{2,2} = 220\ k\Omega}$

R22: https://www.mouser.dk/ProductDetail/Vishay/CRCW0603220KFHEAP?qs=a7EraS9Y3FF6XyVAZTgWrw%3D%3D
R12: https://dk.farnell.com/en-DK/vishay/crcw060339k0fkea/res-39k-1-0-1w-0603-thick-film/dp/2122566

Verify feedback voltage (vref)

> $R_{1,1} := 33E+3 : R_{1,2} := 220E+3 :$

> $R_{1,2} := 68E+3 : R_{2,2} := 39E+3 :$

> $V_{ref1} := V_{out1} \cdot \left(\frac{R_{2,1}}{R_{2,1} + R_{1,1}} \right)$

$$\boldsymbol{V_{ref1} := 1.211881188} \quad (2.1)$$

> $V_{ref2} := V_{out2} \cdot \left(\frac{R_{2,2}}{R_{2,2} + R_{1,2}} \right)$

$$\boldsymbol{V_{ref2} := 1.202803738} \quad (2.2)$$

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